Year 9

REVISION PACK

Unit 1

Name: _____

Class: _____

Unit 1 · Powers of 10 and calculating (Lessons 1.1, 1.2)

Worked example: powers of 10

Work out

(a) 0.46×10^3 (b) $8200 \div 10^4$

(a)	**Step 1** $10^3 = 1000$	$0.46 \times 1000 = 460$
(b)	**Step 1** $10^4 = 10\,000$	$8200 \div 10\,000 = 0.82$

Practice: powers of 10

1.1 Work out

(a) 0.37×10^4 (b) $5600 \div 10^3$

1.2 Work out

(a) 2.08×10^3 (b) $49\,000 \div 10^5$

1.3 Work out

(a) 0.065×10^5 (b) $730 \div 10^4$

1.4 Work out

(a) 1.4×10^2 (b) $26\,000 \div 10^3$

1.5 Work out

(a) 0.009×10^4 (b) $85 \div 10^2$

1.6 Work out

(a) 3.71×10^5 (b) $610\,000 \div 10^6$

Unit 1 · Powers of 10 and calculating (Lessons 1.1, 1.2)

Worked example: estimating

Work out an estimate for $\dfrac{61.4 \times 2.87}{0.32}$. You must show your working.

Step 1	Round to 1 s.f.	$\dfrac{60 \times 3}{0.3}$
Step 2	Top	$60 \times 3 = 180$
Step 3	Divide	$180 \div 0.3 = 600$
Step 4	Answer	600

Practice: estimating

1.7 Work out an estimate for $\dfrac{29.6 \times 4.13}{0.51}$. You must show your working.

1.8 Work out an estimate for $\dfrac{78.2 \times 5.94}{0.19}$. You must show your working.

1.9 Work out an estimate for $\dfrac{9.87 \times 412}{0.23}$. You must show your working.

1.10 Work out an estimate for $\dfrac{21.3 \times 6.8}{0.42}$. You must show your working.

1.11 Work out an estimate for $\dfrac{503 \times 1.96}{0.48}$. You must show your working.

1.12 Work out an estimate for $\dfrac{38.9 \times 7.21}{0.18}$. You must show your working.

Unit 1 · Powers of 10 and calculating (Lessons 1.1, 1.2)

Worked example: using a calculator

Use your calculator to work out $\dfrac{9.6^2 - 4.3^2}{1.8}$

(a) Write down all the figures on your calculator display.

(b) Give your answer to (a) correct to 3 significant figures.

(a)	**Step 1**	Top	$9.6^2 - 4.3^2 = 73.67$
	Step 2	Divide	$73.67 \div 1.8 = 40.92777778$
(b)	**Step 1**	3 significant figures	40.9

Practice: using a calculator

1.13 Use your calculator to work out $\dfrac{11.3^2 - 6.8^2}{2.4}$

(a) Write down all the figures on your calculator display.

(b) Give your answer to (a) correct to 3 significant figures.

1.14 Use your calculator to work out $\dfrac{7.1^2 + 3.6^2}{4.7}$

(a) Write down all the figures on your calculator display.

(b) Give your answer to (a) correct to 3 significant figures.

1.15 Use your calculator to work out $\dfrac{15.2^2 - 9.4^2}{3.3}$

(a) Write down all the figures on your calculator display.

(b) Give your answer to (a) correct to 3 significant figures.

1.16 Use your calculator to work out $\dfrac{8.5^2 - 2.7^2}{1.6}$

(a) Write down all the figures on your calculator display.

(b) Give your answer to (a) correct to 3 significant figures.

1.17 Use your calculator to work out $\dfrac{13.1^2 + 4.2^2}{5.5}$

(a) Write down all the figures on your calculator display.

(b) Give your answer to (a) correct to 3 significant figures.

1.18 Use your calculator to work out $\dfrac{10.6^2 - 7.3^2}{2.9}$

(a) Write down all the figures on your calculator display.

(b) Give your answer to (a) correct to 3 significant figures.

Unit 1 · Indices (Lesson 1.3)

Worked example: index laws

Simplify. Leave your answers in index form.

(a) $4^2 \times 4^6$ (b) $9^8 \div 9^5$ (c) $(6^3)^2$ (d) $\dfrac{3^4 \times 3^5}{3^2}$

(a)	Step 1	Multiply: add the powers	$4^{2+6} = 4^8$
(b)	Step 1	Divide: subtract the powers	$9^{8-5} = 9^3$
(c)	Step 1	Power of a power: multiply	$6^{3\times 2} = 6^6$
(d)	Step 1	Top: add	$3^{4+5} = 3^9$
	Step 2	Divide: subtract	$3^{9-2} = 3^7$

Practice: index laws

2.1 Simplify. Leave your answers in index form.

(a) $7^2 \times 7^5$ (b) $7^{10} \div 7^4$ (c) $(7^5)^3$ (d) $\dfrac{7^3 \times 7^6}{7^4}$

2.2 Simplify. Leave your answers in index form.

(a) $2^6 \times 2^3$ (b) $2^{12} \div 2^7$ (c) $(2^4)^4$ (d) $\dfrac{2^5 \times 2^7}{2^3}$

2.3 Simplify. Leave your answers in index form.

(a) $8^4 \times 8^9$ (b) $8^{11} \div 8^2$ (c) $(8^2)^7$ (d) $\dfrac{8^9 \times 8^2}{8^6}$

2.4 Simplify. Leave your answers in index form.

(a) $5^7 \times 5^2$ (b) $5^9 \div 5^4$ (c) $(5^3)^5$ (d) $\dfrac{5^4 \times 5^4}{5^5}$

2.5 Simplify. Leave your answers in index form.

(a) $6^3 \times 6^8$ (b) $6^{15} \div 6^6$ (c) $(6^7)^2$ (d) $\dfrac{6^8 \times 6^3}{6^9}$

2.6 Simplify. Leave your answers in index form.

(a) $3^5 \times 3^6$ (b) $3^{13} \div 3^8$ (c) $(3^4)^3$ (d) $\dfrac{3^2 \times 3^9}{3^7}$

Worked example: zero and negative powers

Work out the value of

(a) 12^0 (b) 5^{-2}

(a)	**Step 1**	Power 0 is always 1	$12^0 = 1$
(b)	**Step 1**	Negative power: 1 over	$5^{-2} = \frac{1}{5^2}$
	Step 2	Answer	$\frac{1}{25}$

Practice: zero and negative powers

2.7 Work out the value of

(a) 6^0 (b) 3^{-2}

2.8 Work out the value of

(a) 25^0 (b) 2^{-3}

2.9 Work out the value of

(a) 100^0 (b) 10^{-2}

2.10 Work out the value of

(a) 7^0 (b) 3^{-3}

2.11 Work out the value of

(a) 15^0 (b) 6^{-2}

2.12 Work out the value of

(a) 8^0 (b) 2^{-4}

Worked example: a single power and a fraction

This question is about $3^2 \times 3^{-5}$

(a) Write $3^2 \times 3^{-5}$ as a single power of 3 (b) Work out its value as a fraction.

(a)	**Step 1**	Add the powers	$2 + (-5) = -3$
	Step 2	Answer	3^{-3}
(b)	**Step 1**	Negative power: 1 over	$3^{-3} = \frac{1}{3^3}$
	Step 2	Answer	$\frac{1}{27}$

Practice: a single power and a fraction

2.13 This question is about $2^4 \times 2^{-7}$

(a) Write $2^4 \times 2^{-7}$ as a single power of 2 (b) Work out its value as a fraction.

2.14 This question is about $5^1 \times 5^{-3}$

(a) Write $5^1 \times 5^{-3}$ as a single power of 5 (b) Work out its value as a fraction.

2.15 This question is about $4^2 \times 4^{-5}$

(a) Write $4^2 \times 4^{-5}$ as a single power of 4 (b) Work out its value as a fraction.

2.16 This question is about $10^3 \times 10^{-5}$

(a) Write $10^3 \times 10^{-5}$ as a single power of 10 (b) Work out its value as a fraction.

2.17 This question is about $3^1 \times 3^{-4}$

(a) Write $3^1 \times 3^{-4}$ as a single power of 3 (b) Work out its value as a fraction.

2.18 This question is about $2^2 \times 2^{-6}$

(a) Write $2^2 \times 2^{-6}$ as a single power of 2 (b) Work out its value as a fraction.

Unit 1 · Standard form (Lessons 1.4, 1.5)

Worked example: writing numbers in standard form

This question is about standard form.

(a) Write 4.6×10^4 as an ordinary number.

(b) Write 0.0072 in standard form.

(c) Write 830 000 in standard form.

(a)	**Step 1**	$10^4 = 10\,000$	$4.6 \times 10\,000 = 46\,000$
(b)	**Step 1**	Point after the 7	7.2
	Step 2	Small number: negative power	7.2×10^{-3}
(c)	**Step 1**	Point after the 8	8.3
	Step 2	Big number: positive power	8.3×10^5

Practice: writing numbers in standard form

3.1 This question is about standard form.

(a) Write 2.7×10^4 as an ordinary number.

(b) Write 0.00063 in standard form.

(c) Write 8 100 000 in standard form.

3.2 This question is about standard form.

(a) Write 5.04×10^6 as an ordinary number.

(b) Write 0.039 in standard form.

(c) Write 410 000 in standard form.

3.3 This question is about standard form.

(a) Write 9.1×10^3 as an ordinary number.

(b) Write 0.000805 in standard form.

(c) Write 72 000 000 in standard form.

3.4 This question is about standard form.

(a) Write 6.25×10^3 as an ordinary number.

(b) Write 0.0051 in standard form.

(c) Write 940 000 in standard form.

3.5 This question is about standard form.

(a) Write 1.8×10^5 as an ordinary number.

(b) Write 0.00027 in standard form.

(c) Write 3 600 000 in standard form.

3.6 This question is about standard form.

(a) Write 7.3×10^6 as an ordinary number.

(b) Write 0.068 in standard form.

(c) Write 25 000 000 in standard form.

Unit 1 · Standard form (Lessons 1.4, 1.5)

Worked example: multiplying in standard form

Work out $(5 \times 10^4) \times (3 \times 10^6)$. Give your answer in standard form.

Step 1	Multiply the numbers	$5 \times 3 = 15$
Step 2	Add the powers	$10^{4+6} = 10^{10}$
Step 3	15 is more than 10	$15 \times 10^{10} = 1.5 \times 10^{11}$
Step 4	Answer	1.5×10^{11}

Practice: multiplying in standard form

3.7 Work out $(3 \times 10^4) \times (7 \times 10^2)$. Give your answer in standard form.

3.8 Work out $(2.5 \times 10^6) \times (3 \times 10^5)$. Give your answer in standard form.

3.9 Work out $(8 \times 10^3) \times (5 \times 10^7)$. Give your answer in standard form.

3.10 Work out $(1.5 \times 10^9) \times (4 \times 10^2)$. Give your answer in standard form.

3.11 Work out $(6 \times 10^2) \times (9 \times 10^8)$. Give your answer in standard form.

3.12 Work out $(2 \times 10^5) \times (3.5 \times 10^4)$. Give your answer in standard form.

Unit 1 · Standard form (Lessons 1.4, 1.5)

Worked example: dividing in standard form

Work out $(9.6 \times 10^8) \div (3 \times 10^2)$. Give your answer in standard form.

Step 1	Divide the numbers	$9.6 \div 3 = 3.2$
Step 2	Subtract the powers	$10^{8-2} = 10^6$
Step 3	Answer	3.2×10^6

Practice: dividing in standard form

3.13 Work out $(6.3 \times 10^8) \div (3 \times 10^2)$. Give your answer in standard form.

3.14 Work out $(7.5 \times 10^9) \div (2.5 \times 10^4)$. Give your answer in standard form.

3.15 Work out $(4.8 \times 10^6) \div (1.6 \times 10^2)$. Give your answer in standard form.

3.16 Work out $(9 \times 10^7) \div (4.5 \times 10^3)$. Give your answer in standard form.

3.17 Work out $(1.2 \times 10^5) \div (4 \times 10^9)$. Give your answer in standard form.

3.18 Work out $(2.4 \times 10^{10}) \div (8 \times 10^3)$. Give your answer in standard form.

Unit 1 · Standard form (Lessons 1.4, 1.5)

Worked example: ordering

Write these numbers in order of size. Start with the smallest number.

3.1×10^{-2} 8×10^{-3} 2.9×10^{-2} 4.5×10^{-4}

Step 1	Smallest power first	4.5×10^{-4}
Step 2	Next power	8×10^{-3}
Step 3	Same power: compare	2.9×10^{-2}, 3.1×10^{-2}
Step 4	Answer	4.5×10^{-4}, 8×10^{-3}, 2.9×10^{-2}, 3.1×10^{-2}

Practice: ordering

3.19 Write these numbers in order of size. Start with the smallest number.

4.2×10^{-3} 3.9×10^{-2} 7.1×10^{-4} 4.15×10^{-3}

3.20 Write these numbers in order of size. Start with the smallest number.

6×10^{5} 5.8×10^{5} 9.9×10^{4} 1.2×10^{6}

3.21 Write these numbers in order of size. Start with the smallest number.

2.5×10^{-1} 2.45×10^{-1} 9.6×10^{-2} 1.1×10^{-3}

3.22 Write these numbers in order of size. Start with the smallest number.

3.3×10^{-3} 3.03×10^{-3} 1.7×10^{-2} 9.2×10^{-4}

3.23 Write these numbers in order of size. Start with the smallest number.

5.5×10^{4} 5.05×10^{4} 8.1×10^{3} 2.2×10^{5}

3.24 Write these numbers in order of size. Start with the smallest number.

7.4×10^{-5} 1.6×10^{-4} 7.04×10^{-5} 3.9×10^{-6}

Unit 1 · Standard form (Lessons 1.4, 1.5)

Worked example: how many times heavier

The mass of the Sun is 1.99×10^{30} kg. The mass of Venus is 4.87×10^{24} kg. How many times heavier is the Sun than Venus? Give your answer correct to 3 significant figures.

Step 1	Divide	$(1.99 \times 10^{30}) \div (4.87 \times 10^{24})$
Step 2	Calculator	$= 408\,624.2\ldots$
Step 3	Answer	$409\,000$

Practice: how many times heavier

3.25 The mass of Jupiter is 1.9×10^{27} kg. The mass of Mars is 6.42×10^{23} kg. How many times heavier is Jupiter than Mars? Give your answer correct to 3 significant figures.

3.26 The mass of a blue whale is 1.5×10^{5} kg. The mass of a mouse is 2×10^{-2} kg. How many times heavier is a blue whale than a mouse? Give your answer correct to 3 significant figures.

3.27 The mass of Venus is 4.87×10^{24} kg. The mass of Mercury is 3.3×10^{23} kg. How many times heavier is Venus than Mercury? Give your answer correct to 3 significant figures.

3.28 The mass of Saturn is 5.68×10^{26} kg. The mass of Mars is 6.42×10^{23} kg. How many times heavier is Saturn than Mars? Give your answer correct to 3 significant figures.

3.29 The mass of an elephant is 6×10^{3} kg. The mass of an ant is 3×10^{-6} kg. How many times heavier is an elephant than an ant? Give your answer correct to 3 significant figures.

3.30 The mass of Neptune is 1.02×10^{26} kg. The mass of Mercury is 3.3×10^{23} kg. How many times heavier is Neptune than Mercury? Give your answer correct to 3 significant figures.

Unit 1 · Standard form (Lessons 1.4, 1.5)

Worked example: how many fit across

A bacterium is 2.5×10^{-6} m wide. How many of these bacteria would fit end to end across 1 cm? Give your answer in standard form.

Step 1	Change to metres	$1\text{ cm} = 0.01\text{ m}$
Step 2	Divide	$0.01 \div (2.5 \times 10^{-6}) = 4000$
Step 3	Answer	4×10^{3}

Practice: how many fit across

3.31 A cell is 8×10^{-6} m wide. How many of these cells would fit end to end across 2 cm? Give your answer in standard form.

3.32 A grain of sand is 5×10^{-4} m wide. How many of these grains would fit end to end across 4 cm? Give your answer in standard form.

3.33 A hair is 6×10^{-5} m wide. How many of these hairs would fit end to end across 3 cm? Give your answer in standard form.

3.34 An atom is 1.5×10^{-10} m wide. How many of these atoms would fit end to end across 6 mm? Give your answer in standard form.

3.35 A red blood cell is 7×10^{-6} m wide. How many of these cells would fit end to end across 1.4 cm? Give your answer in standard form.

3.36 A virus is 2×10^{-8} m wide. How many of these viruses would fit end to end across 5 cm? Give your answer in standard form.